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Research Article

Microenvironment-Responsive Hydrogel Enclosed with Bioactive Nanoparticle for Synergistic Postoperative Adhesion Prevention

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ABSTRACT: Postoperative adhesion (PA) is a severe complication of abdominal surgery caused by the inability of clinical physical barriers to cope with diverse pathological factors in the process of PA formation. Herein, we described a multifunctional hydrogel composed of bioactive nanoparticles (BNs) and dual-responsive hydrogel to serve as a combination of physical and pharmacological therapy for preventing PA. Specifically, BNs with pro-inflammatory cell-targeted aggregation were designed by integrating hyaluronic acid onto the polydopamine (PDA)-coated hollow ZrO_2 nanoparticles loaded with antimicrobial peptides and platelet lysates that can eliminate bacterial infection and promote tissue repair. PDA can remove the excessive reactive oxygen species (ROS) and thus suppress the oxidative stress damage and accompanying inflammation in the presence of high ROS. The dynamically cross-linked host hydrogel presents injectable yet microenvironment-responsive properties, which enables complete coverage of the uneven tissue and instantly forms a physical barrier to effectively isolate injured tissues and neighboring organs, and synchronously acts as a niche to deliver the BNs in a controlled way. The hydrogel demonstrates a remarkable antiadhesion effect in a rat cecum-abdominal wall adhesion model. Together, this "all-in-one" composite hydrogel strategy capable of a physical barrier capability and pharmacological effects represents a promising clinical solution to prevent PA.

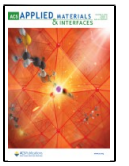


KEYWORDS: *antibacterial, anti-inflammatory, bioactive nanoparticles, composite hydrogel, postoperative adhesion prevention*

1. INTRODUCTION

Postoperative abdominal adhesion (PA) is still a frequent and serious clinical complication after general abdominal surgical procedures, which can be found in up to 90% of patients, with even great improvements in current surgical techniques.[1−3] Once the PA is formed, patients would suffer from severe subsequent consequences including chronic pain, female infertility, intestinal obstruction, and/or organ dysfunction.[4−6] When abdominal adhesions are symptomatic, up to 15−30% of patients indispensably require a reoperative surgery to release the adhesions (i.e., adhesiolysis).[7,8] However, this procedure inevitably causes additional risks including the secondary trauma, hemorrhage, inflammation, and bacterial infection which appear to be even more complicated than primary surgery and lead to the occurrence of more seriously recurrent adhesion.[9,10] Current clinical options for adhesion prevention consist mainly of pharmaceutical administration and physical barrier-based devices.[11−14] Local or systemic administration of anti-inflammatory drugs and anticoagulants (such as aspirin, dexamethasone, heparin, and thrombin inhibitors) is the most frequently employed pharmacological strategy to counteract the adhesion formation.[15,16] In clinical practice, however, such nonspecific therapeutic delivery mechanisms suffer from rapid clearance of drugs in the abdominal cavity as well as interfere with the healing process of injured mesothelium, which collectively results in the compromised therapeutic effects. Rather than using pharmacotherapy, the solid film barrier which are prepared capable of physically isolating the damaged peritoneal surface from other normal tissues or organs until the mesothelial cells are repaired have emerged as a promising and widely used alternative for the prevention of PA. Among which, Interceed (Johnson & Johnson, Cincinnati, USA) and Seprafilm (Genzyme, Cambridge, USA) are the two most common commercial products that have obtained the approval of FDA,[11,17] but are still not ideal in the clinic due to the inconvenient application process as well as the inability to cover and fix on the uneven tissue surfaces. Infusing polymer solutions such as icodextrin (Adept) have been reported to act as liquid physical barriers which could well comply with the

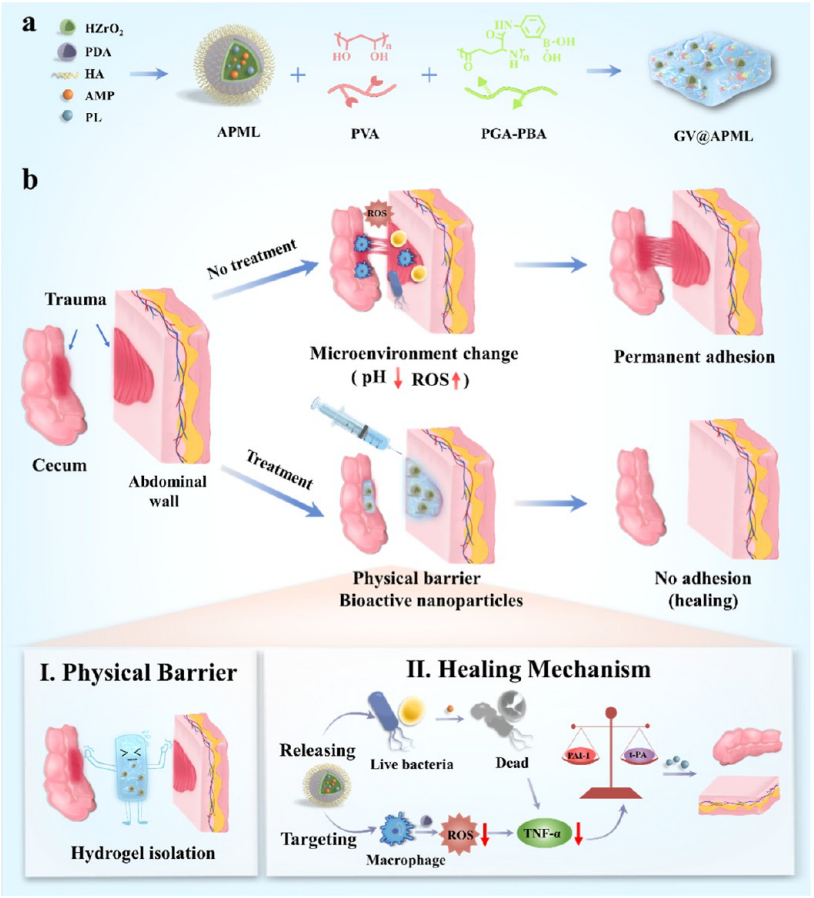
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Scheme 1. (a) Preparation Procedures of a Multifunctional Composite Hydrogel with BNs with Dynamic Borate Ester Bond Cross-Linking Network; (b) Schematic Illustration of a Rat Abdominal Wall-Cecum Adhesion Model with Treatment of Hydrogel Served as a Physical Barrier and Synchronously On-Demand Release Drugs



complex geometry of damaged tissue surface during application,[6,12] while their short residence time at the injured site results in the very limited clinical practice.

More recently, the in situ-formed injectable hydrogels have gained increasing interest for their unique superiority at preventing postoperative adhesions.[18−20] The injectability property allows the hydrogel to be free-flowing and endows them with complete coverage on the injured surface before administration, while in situ forming property could permit the injectable solution transforming into hydrogels on the target position which subsequently serve as a solid barrier locally to physically separate the injured site from the surrounding tissue or organs.[21,22] Combining the advantages of solid film and liquid solution, injectable hydrogels may be promising agents for PA prevention. For example, Stapleton et al.[9] developed a dynamically cross-linked supramolecular polymer−nanoparticle hydrogel equipped with injectability, strong tissue adhesion, and the ability of transforming into viscoelastic barrier in situ, which was able to spray onto tissues as well as vigorous tissue adherence and local retention for 2 weeks in vivo. Zhang et al.[7] designed a viscous, gelatinous yet injectable zwitterionic poly(carboxybetaine acrylamide) (PCBAA) hydrogel which could completely prevent the formation of PA through inhibiting the deposition of fibronectins and fibroblasts that related to the initial and late-stage development of the adhesion, respectively. Despite pioneering advances achieved by the injectable hydrogels, antiadhesive strategies focusing only on physical barrier effect are proved to be insufficient and unsatisfactory to inhibit the formation of PA in more severe and complicated adhesion models due to the complicated pathological factors during the PA occurrence.

Increasing proof indicates that oxidative stress induced by reactive oxygen species (ROS) and the disrupted balance between fibrin formation and fibrinolysis systems play significant roles in promoting peritoneal adhesion.[12,23−26] Particularly, continuous oxidative damage leads to apoptosis and injury of the peritoneal mesothelial cell as well as inhibits normal tissue repair, further promoting inflammation, necrosis, and fibrosis and finally worsening the pathological process. In this context, antioxidant/inflammatory injectable hydrogels aiming at possibly exhausting the ROS and physically separating the injured defect with organs are intensively proposed and were developed recently.[3,27−29] However, those hydrogels eliminate ROS in a nonintelligent and uncontrolled manner which ineluctably influenced on the normal physiological function of ROS. As reported, the appropriate level of ROS and inflammatory factors plays significant roles in promoting wound healing and controlling wound infection which is another common cause of tissue adhesion while it was often overlooked.[30] Hence, the integration of antioxidant hydrogels with tissue-healing and infection-inhibiting properties may get over the limitations of current antioxidant materials and afford a whole course care system to prevent PA. In recent years, therapeutic efficacy of ROS scavenging enabled

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by polydopamine (PDA), because of its rich quinone structure,[31] has aroused great attention in the community of anti-inflammatory biomaterials. Meanwhile, as the first line of defense against microbial infection, antimicrobial peptides (AMP) have been applied to treat bacterial infections due to their potent bactericidal activity.

The supply of exogenous growth factors to injured tissues demonstrates high potential in the field of accelerating healing and tissue regeneration. However, the wound healing experiences complex processes including cell proliferation, migration, angiogenesis, and extracellular matrix deposition, and as a result, requires a specific set of signaling cytokines and growth factors.[32,33] Platelet lysates (PL) containing a plethora of growth promoting factors can be easily obtained from thermal or ultrasound cycles of platelet-rich plasma and have been reported to have potent therapeutic effects on neurogenesis, vascular remodeling, and tissue healing.[34,35] Notably, PL, as an exogenous composite, induce negligible adverse inflammatory effects after administration because the clot and platelet debris are removed during PL processing.[36,37] Nevertheless, the short half-life and easily degraded nature in the proinflammatory proteolytic microenvironment result in the PL and it is difficult to have the desired biological effects in vivo through traditional intravenous administration.

Herein, in order to achieve perfect therapeutic performance and meet the multiple clinical requirements for preventing PA, we designed and developed a multifunctional composite hydrogel (GV@APML) that can quickly cover and adhere onto the irregular injured abdominal wall-cecum surface to serve as a primary physical barrier and synchronously on-demand release bioactive nanoparticles (BNs) to exert multiple pharmacological functions once suffered from endogenic stimulus such as lower pH and higher ROS level (Scheme 1). The BNs (APML) obtained by incorporating hyaluronic acid with pro-inflammatory cell-targeted aggregation onto PDA-coated hollow zirconium dioxide (ZrO_2) nanoparticle loaded with AMP and PL have demonstrated various desirable properties, including targeted drug delivery, ROS scavenging, anti-inflammatory, antibacterial, and wound healing promotion abilities. Notably, the stimuli-responsive drug releasing system combining with the excellent antioxidant and inflammatory response inhibiting properties of PDA construct a favorable microenvironment permitting the optimal utilization of AMP and PL. Meanwhile, the possible side effects arising from the nonintelligent and uncontrolled elimination of ROS can be fitly compensated by the integration of AMP and PL which demonstrate complete eradication of bacterial infection and efficient wound healing promoting capability.

2. EXPERIMENTAL SECTION

2.1. Synthesis and Characterization of HA-PDA@$HZrO_2$@(AMP + PL) (APML) BNs. *2.1.1. Synthesis of Hollow ZrO_2 Nanoliveparticles.* The SiO_2 nanoparticles as the templates for preparing hollow ZrO_2 ($HZrO_2$) nanoparticles were first synthesized through the hydrolysis of tetraethyl orthosilicate (TEOS). Specifically, 15 mL of TEOS was added in a mixed solution containing ammonium hydroxide (7.5 mL), ethyl alcohol (350 mL), and water (25 mL) and stirred for 6 h. Thereafter, the mixed solution was washed with methanol to remove excess unreactive substances to acquire SiO_2 nanoparticles. Then the acquired SiO_2 was redispersed in ethanol and acetonitrile mixture (3:1) and ammonium hydroxide and zirconium-(IV) propoxide solution were added and stirred for 10 h. The reagents were washed with ultrapure water by centrifugation (11,000 rpm, 8 min) to remove free zirconium(IV) propoxide and then the ZrO_2@SiO_2 were obtained. To obtain the hollow zirconium dioxide ($HZrO_2$) nanoparticles, the ZrO_2@SiO_2 was redispersed in sodium hydroxide solution (0.1 g/mL) and stirred for 12 h at 60 °C. After washing with ultrapure water and freeze-drying, the $HZrO_2$ nanoparticles were collected.

2.1.2. Synthesis of HA-PDA@$HZrO_2$ (APZ) Nanoparticles. $HZrO_2$ and DA (equal mass equivalents) were dispersed in Tris buffer (pH = 8.5) and stirred for 48 h, after which the samples were washed three times with ultrapure water to obtain PDA@$HZrO_2$ (PZ). Subsequently, 0.1 g of NH_2−HA was added into the fresh Tris buffer (10 mL, pH = 8.5) containing 0.1 g of PDA@$HZrO_2$ and stirred at room temperature for 24 h. The obtained HA-PDA@$HZrO_2$ (APZ) was purified by centrifugation (11,000 rpm, 8 min) against water and collected after lyophilization.

2.1.3. Synthesis of HA-PDA@$HZrO_2$@(AMP + PL) (APML) BNs. For loading of the AMP and PL, APZ (100 mg) was dispersed in 20 mL of mixed water solution containing 50 mg of AMP and 200 μL of PL, followed by continuous stirring for 12 h in vacuum. Then, the supernatant of the mixture was removed by centrifugation. Thus, the HA-PDA@$HZrO_2$@(AMP + PL) (APML) BNs were successfully prepared.

2.2. Preparation and Characterization of PGA−PBA/PVA@APML (GV@APML) Hydrogels. *2.2.1. Synthesis and Characterization of PGA−PBA.* To synthesize PGA−PBA, *N*-(3-(dimethylamino)propyl)-*N*′-ethylcarbodiimide hydrochloride (EDC) and *N*-hydroxysuccinimide (NHS) act as coupling agents to conjugate 3-aminophenylboronic acid (PBA) to poly(γ-glutamic acid) (γ-PGA). In a typical reaction, γ-PGA (1 g) was dissolved in 50 mL of MES buffer (pH 5.5), followed by the addition of EDC (two mol equiv of carboxyl group in γ-PGA) and NHS (one mol equiv of carboxyl group in γ-PGA) to activate the carboxyl groups in γ-PGA for 2 h. Then, PBA (two mol equiv of carboxyl group in γ-PGA) was added into the reaction mixture and stirred for another 48 h at room temperature. Subsequently, the PGA−PBA solution was dialyzed in ultrapure water for 3 days and then freeze-dried.

2.2.2. Preparation of GV@APML Hydrogels. PGA−PBA solution (pH 7.4, PBS) with concentration 4% w/v was added dropwise to 2% w/v PVA solution (pH 7.4, PBS) containing 10% w/v APML with constant stirring. The drug-loaded GV@APML hydrogels immediately formed in seconds. APML-free hydrogels (GV) were successfully prepared by using the same method. After lyophilization, the hydrogels fractured brittlely in liquid nitrogen and were sputter-coated with gold for scanning electron microscope (SEM) analysis.

2.2.3. Viscosity and Rheological Characterization. The rheological properties of hydrogels were evaluated by employing a rheometer. The storage modulus (G') and loss modulus (G'') were investigated at 37 °C. A time sweep test was conducted at a 1% strain amplitude and a frequency of 1 rad s^{-1}. A shear-thinning test was conducted at 25 °C with 1 Hz frequency and 1.0% strain to evaluate the injectable property.

2.2.4. Response of the GV Hydrogel to pH and ROS Change. The obtained GV hydrogel was put into centrifuge tubes, and 200 μL of acidic PBS buffer (pH5.5) or H_2O_2 (10 mM) was added on the hydrogel. After that, the hydrogel's status was recorded.

2.2.5. Degradation Test. In vitro degradation behavior of the hydrogels was conducted by immersing the lyophilized samples into centrifuge tubes of different conditions (pH = 7.4, pH = 5.5 or with H_2O_2). The weight of each sample was weighted at multiple time points (0, 2, 4, 6, 8, 12, and 24 h), and the degradation ratio was calculated.

2.3. Cellular Uptake. LPS-activated macrophages (RAW264.7 cells) were incubated with FITC-conjugated PZ or APZ (100 μg/mL) for 3 h. After that, macrophages were stained with DAPI for 10 min to detect the fluorescence images.

2.4. ROS-Scavenging Ability Evaluation. The antioxidant activity evaluation of nanoparticles was performed by scavenging free radical DPPH. Samples with different concentrations were mixed with DPPH to react for a certain time. Then, the full wavenumber scanning curves were recorded using a UV−vis spectrophotometer.

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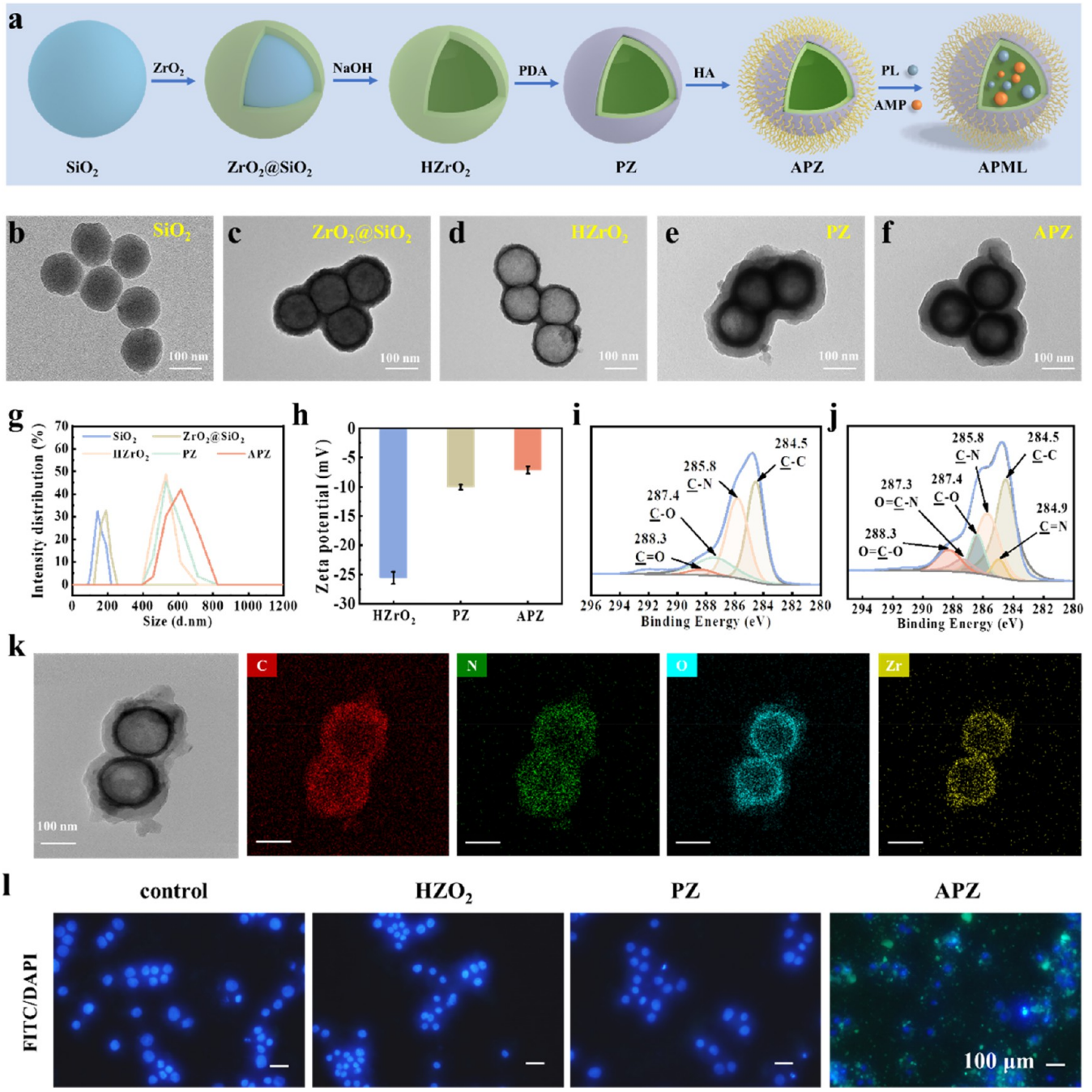


Figure 1. Characterization of APML nanoparticles. (a) Schematic diagram for the preparation of APML BNs. TEM images of various nanoparticles including SiO_2 (b), ZrO_2@SiO_2 (c), $HZrO_2$ (d), PZ (e), and APZ (f). (g) The size distribution of various nanoparticles. (h) Zeta potential of various nanoparticles. XPS of PZ (i) and APZ (j). (k) Elemental mappings of C, N, O, and Zr of APML. (l) Macrophage-targeting capability of APZ. Representative fluorescence images of macrophages incubated with different FITC-samples

2.5. In Vitro Release Profiles of PL and AMP. The nanoparticles were added to PBS (pH 7.4) and placed on a shaker to simulate drug release. For the AMP release assay, a standard curve was established by the UV−vis spectrum test. The APML nanoparticles were immersed in 1 mL PBS (pH7.4) at 37 °C and shaken at 120 rpm. At regular intervals, 1 mL of the supernatant was withdrawn from the release media and the cumulative amount of AMP released from nanoparticles was determined by UV−vis analysis.

Similarly, supernatants were collected at fixed time points, and the concentration of PL in the supernatant was quantitatively detected using a BCA protein assay kit. The release content and ratio of each test were recorded to draw a PL release curve.

2.6. Intracellular ROS-Scavenging Evaluation. In vitro ROS scavenging ability of the nanoparticles in RAW264.7 cells was detected with DCFH-DA probe by a standard procedure. In short, RAW264.7 cells were seeded on 48-well plates and treated with different nanoparticles containing 100 ng/mL LPS, while the control group was only treated with LPS. After treatment for 24 h, cells were stained with DCFH-DA (30 min) and DAPI (10 min), respectively.

2.7. In Vitro Antibacterial Experiments. In vitro antibacterial activity of nanoparticles was assessed by using *Staphylococcus aureus* and *Escherichia coli*. Bacterial suspensions in the exponential growth phase were diluted to 1×10^5 CFU/mL. The nanoparticles and sterile PBS as control group were incubated with the bacterial solution, respectively, at 37 °C. Subsequently, 100 μL of bacterial solution was

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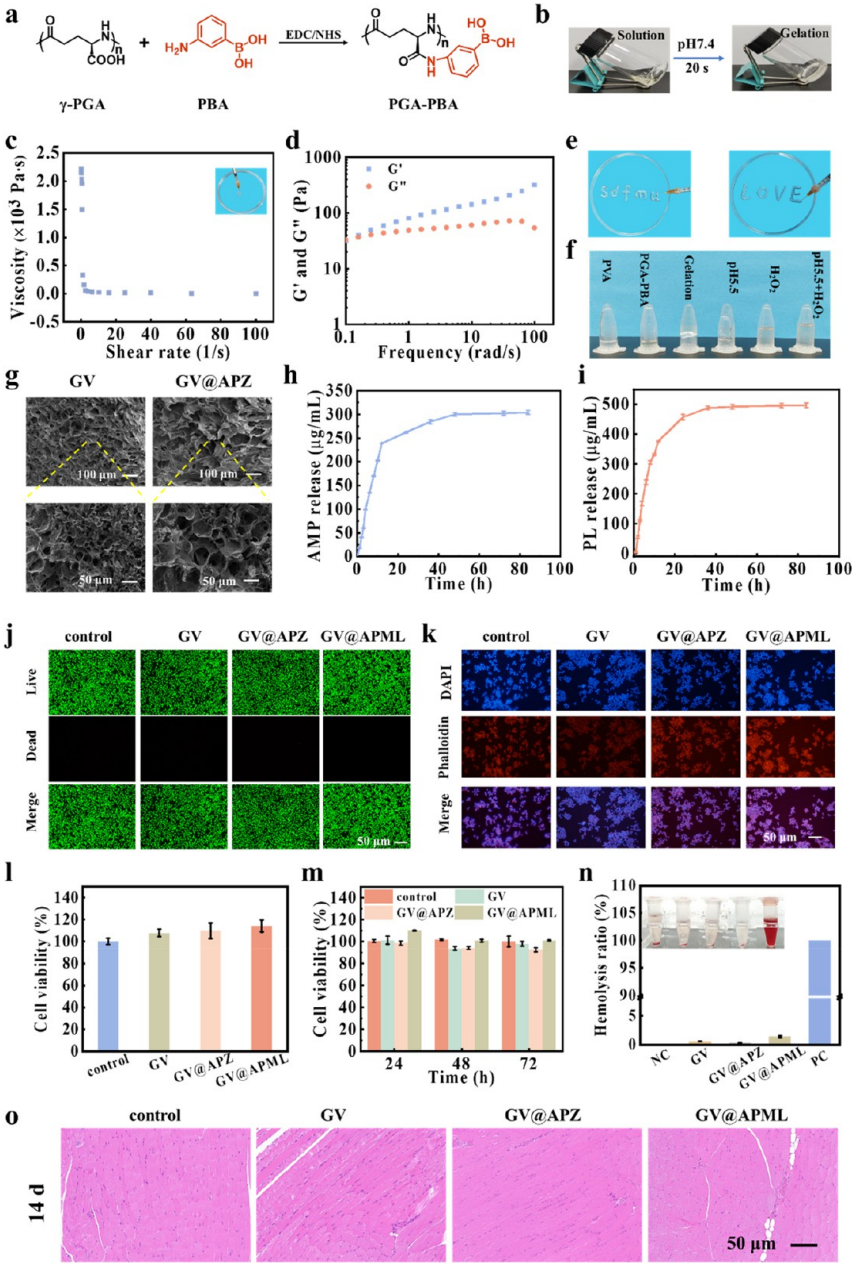


Figure 2. Characterization of hydrogels and drug release in vitro. (a) A synthetic route of PGA−PBA polymer. (b) Gelation of GV hydrogel. (c) The viscosity of GV hydrogel at increasing shear rate. (d) Rheological analysis of GV hydrogel in frequency sweep mode. (e) Injectability of GV hydrogels. (f) The images of the hydrogel to pH and H_2O_2 changes. The hydrogel was placed in acidic PBS buffer (pH 5.5) or 10 mM H_2O_2 for 6 h, the hydrogel's shape was recorded. (g) The SEM for hydrogels after freeze-drying. The AMP (h) and PL (i) cumulative release at different time points. (j) Representative live/dead images of L929 cells cultured with different samples for 24 h. (k) Representative morphological images of RAW264.7 cells by cytoskeleton staining. Quantitative analysis of CCK-8 assay for L929 cells (l) and RAW264.7 cells (m). (n) Hemolysis ratio of different hydrogels (NC and PC represent negative control and positive control). (o) H&E staining of the muscle tissues from mice on day 14 after treatment.

seeded on agar plate to give visible colonies for counting analysis. Morphologies of bacteria were also used to test the antibacterial ability of nanoparticles. Similarly, the bacteria were incubated with different samples, and then the supernatant was collected after centrifugation. The bacterial suspension was dripped onto silicon wafers and immersed in a 4% glutaraldehyde solution and the adherent bacteria were characterized by a SEM.

2.8. Cell Scratch Experiment. 2 ×10^5 L929 cells were seeded in 24-well plates to achieve more than 90% coverage. After that, a scratch of the same width was scratched out at the bottom of each well. Next, the hydrogel extraction was added into the corresponding wells for coincubation. At certain culture times, the area of cell migration was recorded.

2.9. Biocompatibility Test. Red blood cells (RBCs) were separated from the serum by centrifugation (1000 rpm, 15 min). RBCs treated with ultrapure water and normal saline were used as positive (PC) and negative (NC) controls, respectively. Equal volumes of RBC suspension and hydrogel were mixed and incubated at 37 °C and the upper supernatant was collected by centrifugation to measure the absorbance at 540 nm.

CCK-8 assay, the live/dead fluorescent staining, and cytoskeleton staining were used to measure the cell viability. Briefly, L929 fibroblast cells were seeded in a 96-well plate, and extract of the

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hydrogel was added into each well to incubate for certain time intervals. Cell viabilities were detected by the CCK-8 assay. Furthermore, live/dead staining was performed. After being incubated with the hydrogel extract, the cells were stained with CalceinAM/propidium iodide dye, and the green and red fluorescence of the cells were observed under a fluorescence microscope. Additionally, the morphological changes in RAW264.7 cells induced by hydrogels were observed by cytoskeleton staining. The cytoskeleton of RAW264.7 cells was stained with TRITC-phalloidin, and the nucleus was stained with DAPI. The morphology of the macrophages was observed by fluorescence microscopy.

To further evaluate the biocompatibility of the hydrogel, histological evaluation was used to assess toxicity in vivo. Female BALB/c mice were used in this study under protocols approved by the Committee for Animal Experimental Ethical Inspection of Shandong First Medical University. The freeze-dried hydrogel samples were implanted subcutaneously on the back of mice. The implanted specimens and surrounding tissues were removed after normal feeding for 7 and 14 days. The back muscles were then dissected and subjected to histological analysis.

2.10. Anti-infectious Assessment In Vivo. The model of abdominal wall defect infection was established in Sprague−Dawley (SD) rats. Based on our previous work,[38] the in vivo anti-infectious evaluation of hydrogel was conducted. The peritoneum was dissected to create a 1 × 1 cm^2 abdominal wall defect, and then 200 μL (10^6 CFU mL^{-1} LB medium) of *S. aureus* suspension was added. The wounds were covered with samples in hydrogel group, while there was no treatment in the control group. All experimental procedures were carried out under sterile conditions. The rats were euthanized on day 3. The tissues were harvested for antibacterial assay and histological analysis. H&E, CD68, and IL-6 immunohistochemical staining were used to detect inflammation at the wound site.

2.11. In Vivo Antiadhesion Test. Sidewall defect-cecum abrasion in a SD rat model was performed according to previous literature reports.[6,39] Female SD rats were divided into three groups with 10 rats in each group. Briefly, the cecum was rubbed with sterile surgical gauze until punctate bleeding appeared on the surface of the cecum to prepare the cecum defect. Next, a full-thickness abdominal wall injury and abrasion of the cecum were constructed by using a surgical scissor and gauzes on a rat model. The control group was washed only with 1 mL PBS, and the hydrogel group were coated with 1 mL hydrogel on the injured surfaces. On postoperative days 7 and 14, the rats in each group were subjected to gross observation and the adhesion score was evaluated according to the standard adhesion scoring system (Table S1). All defect tissues were explanted for histologic examination. The repaired areas were photographed and then stained with H&E staining, Masson trichrome staining, immunohistochemistry staining of CD68, CD80, tumor necrosis factor-α (TNF-α), and immunofluorescence staining of α-smooth muscle actin (α-SMA). To investigate the expression of PAI-1 and tissue-type plasminogen activator (t-PA) in tissues related to adhesion, PAI-1 and t-PA were detected. The results of histological images were acquired by a fluorescence microscope and quantitatively analyzed by ImageJ software.

3. RESULTS AND DISCUSSION

3.1. Preparation and Characterization of APML Nanoparticles. The process to prepare the BNs (APML) is illustrated in Figure 1a. First, the $HZrO_2$ spheres were prepared by a sacrificing template method as previously reported. As revealed by SEM and transmission electron microscopy (TEM) images (Figures S1 and 1b), the prepared template (SiO_2) exhibited a typical spherical appearance with an average diameter of approximately 110 nm. After the introduction of ZrO_2, the ZrO_2@SiO_2 displayed the uniform core−shell structure and the average diameter increased to about 142 nm (Figures S1 and 1c). The $HZrO_2$ nanoshells could be readily obtained via template etching by a NaOH solution (Figures S1 and 1d). Subsequently, PDA was introduced to coat onto the surface of $HZrO_2$ to serve as both ROS removers and vehicle for further functionalization. As shown in Figures S1 and 1e, the PDA@$HZrO_2$ nanoparticles (denoted as PZ) demonstrated obvious hierarchical structures with a sharp increase of diameter from ∼142 to ∼207 nm. Taking advantages of the Schiff base or Michael addition reactions between PDA and amine- or thiol-containing molecules,[40−42] NH_2−HA was easily to be covalently immobilized on the PZ, and the corresponding sample was denoted as APZ (Figures 1f and S1). In these processes, the hydrodynamic sizes of the nanoparticles were also monitored by dynamic light scattering and all of them showed satisfactory monodispersity (Figure 1g). Additionally, the surface potential of APZ was increased to −7.7 mV after modification with NH_2−HA compared with PZ (−10.0 mV), indicating the change of the surface chemical composition of the nanoparticles (Figure 1h). The chemistry of PZ and APZ were explored through X-ray photoelectron spectroscopy (XPS). To further explore the relationship between surface properties and composition, the chemical compositions of PZ and APZ were characterized by XPS. In contrast to PZ (Figure 1i), functional groups of −C═N− at 283.98 eV in APZ (Figure 1j) appeared and confirmed that NH_2−HA was introduced to the surface of PZ particles through Schiff base bonds. To examine the distribution pattern of AMP and PL in APZ nanoparticles, energy dispersive spectroscopy and mapping were employed. The elemental composition (C, N, O, and Zr) of APML can be directly observed through the element mapping in Figure 1k, which confirms the formation of APML nanoparticles. Taken together, these results demonstrated that hollow APML nanoparticles were obtained successfully.

3.2. Synthesis and Characterization of the Polymer PGA−PBA. Due to the biocompatibility, biodegradability, and structural suitability with an abundance of carboxyl repeat units, γ-PGA was selected as the backbone material of the hydrogel. The carboxyl group was first activated by EDC and NHS, and then PBA was introduced into the γ-PGA molecular backbone and the resulting sample was denoted as PGA−PBA (Figure 2a). For confirming the successful immobilization of the PBA moiety, the PGA−PBA was further characterized by NMR spectroscopy. As exhibited by the ^{1}H NMR spectra in Figure S2a, the protons of the phenyl group showed peaks at 7.0−7.7 ppm, and the characteristic peaks at 4.2 ppm belonged to the protons in the methyl unit [−COOH (NH_2)−] of the γ-PGA backbone, which validated the conjugation of PBA with γ-PGA. The PBA grafting degree for γ-PGA calculated by 1H NMR was determined to be 17.8%. To further verify the above results, changes in chemical structure were detected using Fourier transform infrared (FTIR) spectroscopy. As depicted in Figure S2b, the peaks were at 813, 775, and 771 cm^{-1}, representing the bending vibration of the benzene ring. Additionally, the newly appeared peaks at 1660 and 1389 cm^{-1} corresponded to amide I band and amide II band, respectively, suggesting that PBA was successfully conjugated to γ-PGA. Figure S2c shows the UV visible spectrum of the PBA−PGA polymer. PBA−PGA polymer with a maximum absorption peak at 295 nm was similar to PBA, whereas γ-PGA has no absorption peak. Collectively, these results indicated PGA−PBA polymer had been successfully synthesized, which can be used for the preparation of hydrogel encapsulating drug nanoparticles.

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ACS Appl. Mater. Interfaces 2024, 16, 60933−60947

3.3. Fabrication and Characterization of the PGA−PBA/PVA (GV) Hydrogel. Stimuli-responsive hydrogels have gained much attention to achieve controlled and rapid drug release at the specific site, which further improves the bioavailability of drugs and enhances the treatment efficiency. Herein, taking full advantage of the sensitivity of dynamic borate ester bonds to pH and ROS, the as prepared PGA−PBA as the backbone material was cross-linked with poly(vinyl alcohol) (PVA) to construct the multiresponsive hydrogel (GV) through the formation of phenylboronic acid−diol ester bonds. As shown in Figure 2b, the GV hydrogel could be rapidly formed (within 20 s) at a physiological pH value. Additionally, the dynamic formation−dissociation cycles of reversible borate ester bond within the GV hydrogels constantly happened at a molecular level,[43] which could result in a lot of unusual properties, such as injectable and self-healing. In this context, to assess the injectability of hydrogels, the viscosities were tested with increasing shear rates, and results are shown in Figure 2c. With the increase of shear rate, the viscosity of the hydrogel decreased rapidly, reaching about 6 Pa·s at 60 s^{-1}. Such shear-thinning behavior indicated the distinct injectable property of GV hydrogel, which was further vividly demonstrated by injecting the hydrogels through a 1 mL syringe (Figure 2c inset image). As depicted in Figure 2e, the hydrogel can be easily injected into a 1 mL syringe needle to shape into "LOVE and sdfmu". Apart from stimuli-responsive property, dynamic bonds cross-linked hydrogels usually showed promising properties, including perfect biological tissues matching mechanical properties, and self-healing performance for extending their applications. Hence, a rheometer was used to analyze the rheological properties of the hydrogel. In rheological analysis, the gel point was defined as the intersection of storage modulus (G') and loss modulus (G''), that is, the critical state between solid and liquid. As revealed in Figure 2d, the hydrogel formation of GV hydrogel was proved by dynamic frequency-sweep tests. From high frequency to low frequency, G'' was less than G' for the GV hydrogel, exhibiting a gel-like character. The self-healing process was also assessed through cutting the intact hydrogel in half. As exhibited in Figure S3, the different parts could instantly form into one whole hydrogel after recontact to completely cover and defend the damaged tissue surface, showing the remarkable self-healing property for the constructed hydrogel. Subsequently, the duel-responsive property of the GV hydrogel was evaluated at lower pH values or in the presence of abundant ROS, which simulated the postoperative wound microenvironment. As shown in Figure 2f, the hydrogel collapsed both at lower pH (pH = 5.5) and higher ROS (H_2O_2,10 mM) conditions owing to the reversible boronic ester bond. The dual responsive properties of the hydrogel were further verified in the degradation curve of the hydrogel (Figure S4). Additionally, the GV@APML hydrogels consisting of APML nanoparticles were synthesized by controlling the concentrations of nanoparticle components in this study. The homogeneously porous structures were observed in both the GV and nanoparticle-loaded hydrogels through SEM (Figure 2g), which indicated that the addition of nanoparticles did not change the original structure of the hydrogel.

3.4. Drug Release Analysis. The AMP release curves over time was determined using a standard calibration curve obtained from the typical UV−vis absorption peak of AMP at 280 nm. As exhibited by the sustained release curve of AMP in Figure 2h, APML showed a rapid release before 12 h, followed by a slow release process with the releasing amount of 304.3 μg/mL after 84 h. Further, the concentration of PL was measured though the BCA protein assay kit. During the first 12 h, PL showed a rapid release trend similar to that of AMP, and a final release of 496.4 μg/mL was detected (Figure 2i). The results revealed that AMP and PL could be readily released from the as-prepared APML by a simple diffusion.

3.5. Evaluation of the Biocompatibility of the Composite Hydrogels. Biocompatibility is one of the most important prerequisites for identifying the latent risks and safety issues in new development of biomedical materials.[44−46] Thus, the cytocompatibility and hemocompatibility of the GV@APML hydrogel were assessed in vitro. Testing the cytotoxicity to L929 cells or RAW264.7 cells by the live/dead staining, cytoskeleton staining (DAPI and TRITC-phalloidin), and CCK-8 assay to assess the cytocompatibility of the GV@APML. L929 cells of the control group and sample groups showed green fluorescence (live cells), and almost no red fluorescence appeared (dead cells) in Figure 2j after incubation with the various samples for 24 h. Figure S8 displayed that the L929 cells in samples possessed a typical spindle-like shape as well as similar amounts of adhered cells and cell spreading. The CCK-8 assay also illustrated that the L929 cell viabilities of each group were >100% (Figure 2l). As exhibited in Figure 2k,m, the cytoskeleton staining and CCK-8 assay results of RAW264.7 cells treated with the different samples were the same as the results of L929 cells. Above results collectively indicated that the GV@APML had excellent cytocompatibility promising its potential application in PA prevention. Moreover, the hemolysis test was used to measure the hemocompatibility of GV@APML hydrogel. The samples for the negative control group (NC) and positive control group (PC) were normal saline and ultrapure water, respectively. The suspension of each sample group was a yellowish color or light red in Figure 2n, which was the same as in the normal saline-treated group. Whereas, the suspension of the ultrapure group appeared deep red with no sediment at the bottom of the tube. Additionally, it was obviously showed that the hemolysis ratios of each sample group were not higher than 5%, which was considered a safe level of hemostatic materials.[47] The results implied that GV@APML possessed outstanding hemocompatibility.

To further assess the biocompatibility in vivo, the hydrogels were, respectively, implanted subcutaneously on the back of mice for 7 and 14 days. The mouse skin tissues getting in touch with the hydrogel samples were taken on the seventh and 14th day, respectively, and the untreated mouse skin tissues were used as the control. Hematoxylin−eosin (H&E) staining was performed on all skin tissues for further histological analysis. No obvious inflammatory lesions or tissue damage were observed in the muscle tissues of any groups in Figures S5 and S2o, further confirming the excellent in vivo biosafety of the constructed GV@APML system. In summary, the above results indicated that the GV@APML hydrogel had remarkable biocompatibility both in vitro and in vivo and will not cause distinct inflammatory response.

3.6. Evaluation of the Macrophage Targeting Capability and ROS-Scavenging. Targeted delivery of drugs is increasingly developed to reduce undesirable side effects in normal cells. Inflammatory-targeting drug delivery system based on the specific recognition of HA for CD44 receptor overexpressed in a variety of inflammatory cells is a promising

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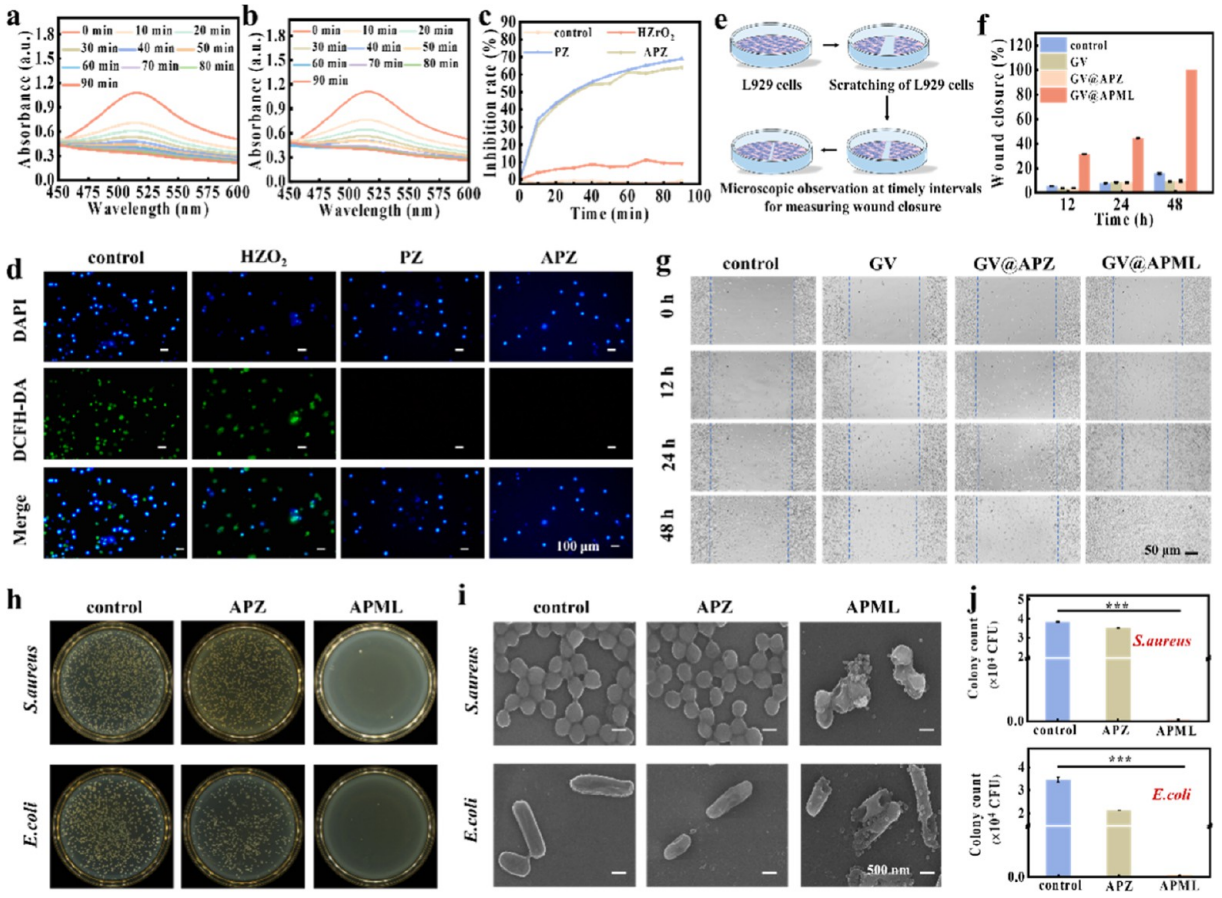


Figure 3. DPPH scavenging efficiency over time for PZ (a), APZ (b). (c) DPPH scavenging percentage by nanoparticles. (d) The reduction of oxidative stress in RAW264.7 cells was monitored by the DCFH-DA for various treatment. (e) Schematic illustration of cell scratch experiment. (f) Quantification of L929 cells migration on different samples. (g) Images of wound scratch migration assay at different times. Evaluation of the antibacterial effect in vitro. (h) Images of survival bacteria colonies grown on agar plates for *S. aureus* and *E. coli*. (i) The morphology of *S. aureus* and *E. coli* after various treatment by SEM. (j) CFU counts of bacteria with various treatment.

mode of precise therapy.[48−51] In this study, the targeting properties of HA modified PZ nanoparticles (APZ) were studied in vitro through LPS-activated proinflammatory macrophages as a model cell. To visualize the distribution around the cells, the PZ and APZ nanoparticles were first labeled by the fluorescent FITC. After incubation with FITC-APZ or FITC-PZ for 4 h, the fluorescence condition of macrophages was measured by inverted fluorescence microscope. PZ-treated macrophages exhibited no green fluorescence, whereas a distinctly green fluorescence signal was observed in the HA conjugated APZ-treated macrophage group in Figure 1l. To confirm this observation, TEM images were used to directly detect the distribution of nanoparticles. As shown in Figure S6, there were many more particles existing in the macrophages incubated with APZ than those processed with PZ. The results collectively indicated that APZ nanoparticles could specifically combine with activated macrophages, which will lay the foundation for the further efficient and safe application of drugs.

Excessive ROS at the abdominal wound sites usually leads to strong oxidative stress, delayed wound healing, and ultimately even promotion of the formation of severe postoperative adhesions. Hence, the ROS-scavenging capability of materials is indispensable to prevent PA. In this context, the antioxidative property of the BNs was evaluated using DPPH serving as a ROS-specific probe that the absorbance of the DPPH solution at 517 nm could be reduced by the single electron pairing provided by antioxidants. As indicated in Figures S7a,b and 3a−c, no remarkable absorbance variations were detected in the control and $HZrO_2$ group, while the absorption intensity of PZ and APZ groups gradually decreased over time, and ultimately both PZ and APZ nanoparticle-treated groups could achieve more than 70% scavenging rate of the free radical compared to control and $HZrO_2$ groups, demonstrating that the integration of PDA could substantially increase the ability of DPPH scavenging. Furthermore, the intracellular ROS-scavenging ability of nanoparticles was tested by means of LPS (an inducer for inflammatory cells producing ROS)-stimulated RAW264.7 cell as a model. Specially, nonfluorescent DCFH-DA which could be converted into highly fluorescent 2′,7′-dichlorofluorescein (DCF) in the presence of ROS was used as a probe to detect the intracellular ROS level. As shown in Figure 3d, with comparison to control and $HZrO_2$, PZ and APZ groups containing PDA integration displayed no green fluorescence, indicating that the existence of PDA rendered the constructed composite particles remarkable ROS scavenging ability.

3.7. Cell Scratch Test of the Hydrogels. The migration and growth of cells are fundamental requirements for tissue repair and regeneration. Hence, a scratch wound was created in a monolayer of L929 fibroblasts for investigating the effect of GV@APML on cell migration (Figure 3e). The scratch results

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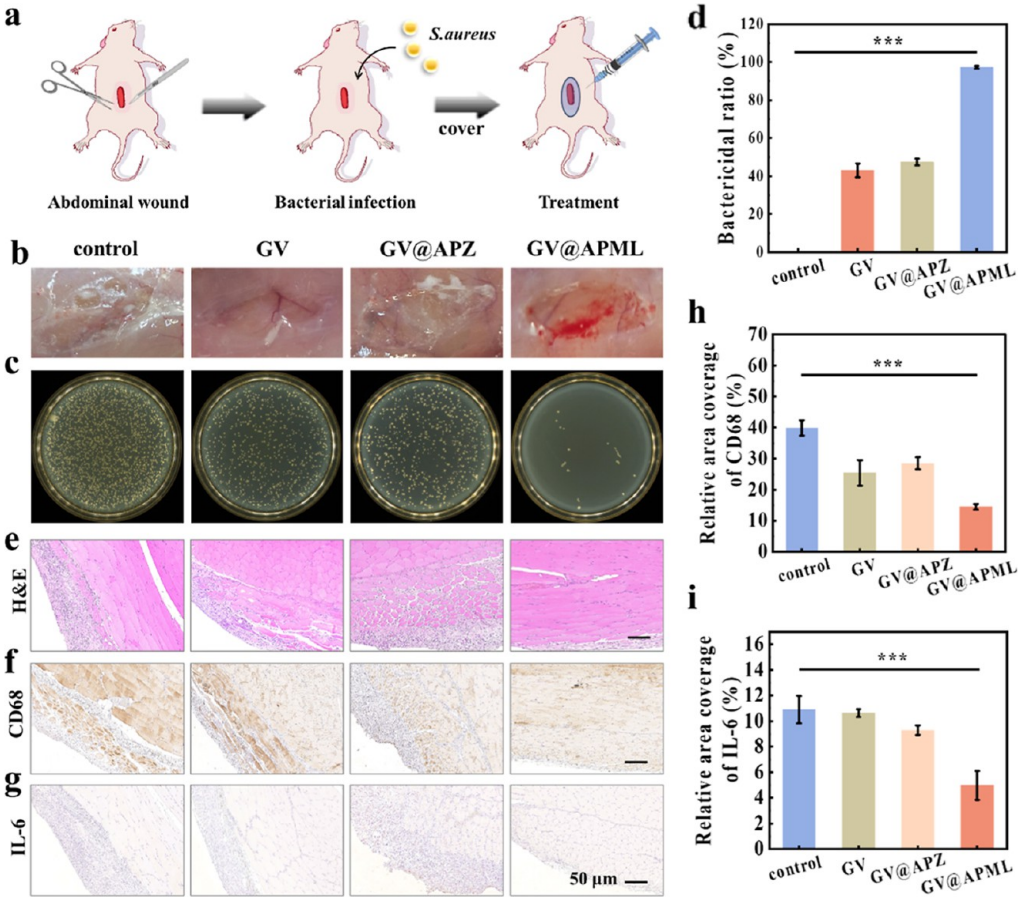


Figure 4. Analysis of the effect of hydrogels in treating infected abdominal wounds in vivo. (a) Schematic illustration showing the overall processes of in vivo antibacterial experiments. (b) Representative images of wounds by treatment with different samples. (c) Representative images of bacterial colony at wound sites. (d) Bactericidal ratio statistics. (e) Representative images of H&E staining of the wounds. (f) Immunohistochemistry staining and (h) qualitative analysis of CD68. (g) Immunohistochemistry staining and (i) qualitative analysis of IL-6.

of cells were recorded at 0, 12, 24, and 48 h for calculating the rate of wound closure. As shown in Figure 3g, the cells migrated faster and filled in gaps in the GV@APML group after 48 h of incubation. The quantitative statistic showed that among the three hydrogel groups, the GV@APML group had the greatest effect on cell migration and proliferation, achieving a scratch healing rate of nearly 100% at 48 h, while only 10% of the wound-healing ratio could be achieved by the GV@APZ group (Figure 3f). The results suggested that GV@APML could promote cell proliferation and migration mainly related to the released PL which has been well proved to be able to exert potent therapeutic effects on neurogenesis, vascular remodeling, and tissue repair.[36,52] Therefore, the composite GV@APML hydrogel might be a promising option for abdominal injury repair.

3.8. Evaluation of the Antibacterial Activity In Vitro. Bacterial contamination is one of the prime reasons of unhealing abdominal injury, which also prolongs the inflammatory process and increases the risk of PA.[53−56] Thus, *S. aureus* and *E. coli* were chosen as representative Gram-negative and Gram-positive strains to assess the antibacterial performance in vitro. To quantitatively assess the bactericidal capability of samples, a plate colony counting method was conducted. As depicted in Figure 3h,j, there was no obvious difference of the colony counts in the APZ group and the control for both *S. aureus* and *E. coli*, indicating the nanoparticles had no bactericidal effect for the bacterial cells in the absence of AMP loading. By comparison, the bactericidal efficiency of APML could achieve a very high rate (99.7% for *S. aureus* and 99.5% for *E. coli*), suggesting that the hybrid nanoparticle could perform remarkable and broad-spectrum bactericidal performances after the introduction of AMP. To further explore the bactericidal mechanisms, changes in bacterial morphology were observed by SEM. As shown in Figure 3i, the bacterial cells processed with either PBS or APZ group presented full and intact appearances presenting typical identifications of live bacteria. In contrast, severely destroyed bacterial membranes had occurred in both bacterial cell types after treating with the APML group, confirming the powerful membrane breaking effect of AMP released from the APML particles. Collectively, the constructed APML particles could perform remarkable antibacterial property toward both *S. aureus* and *E. coli* in vitro.

3.9. In Vivo Antibacterial and Anti-inflammatory Properties of Hydrogels. Bacterial infection leads to increased inflammation at the wound site and further aggravates the development of adhesions.[11] The above results suggest that the components of APML nanoparticles have efficient bactericidal and ROS scavenging abilities in vitro. To further investigate the in vivo performances for the constructed composite hydrogel composed of GV and APML, the *S. aureus*-infected abdominal wall wound model was established in SD rats and processed as illustrated in Figure 4a. The *S. aureus*-infected abdominal wall wounds were treated with PBS

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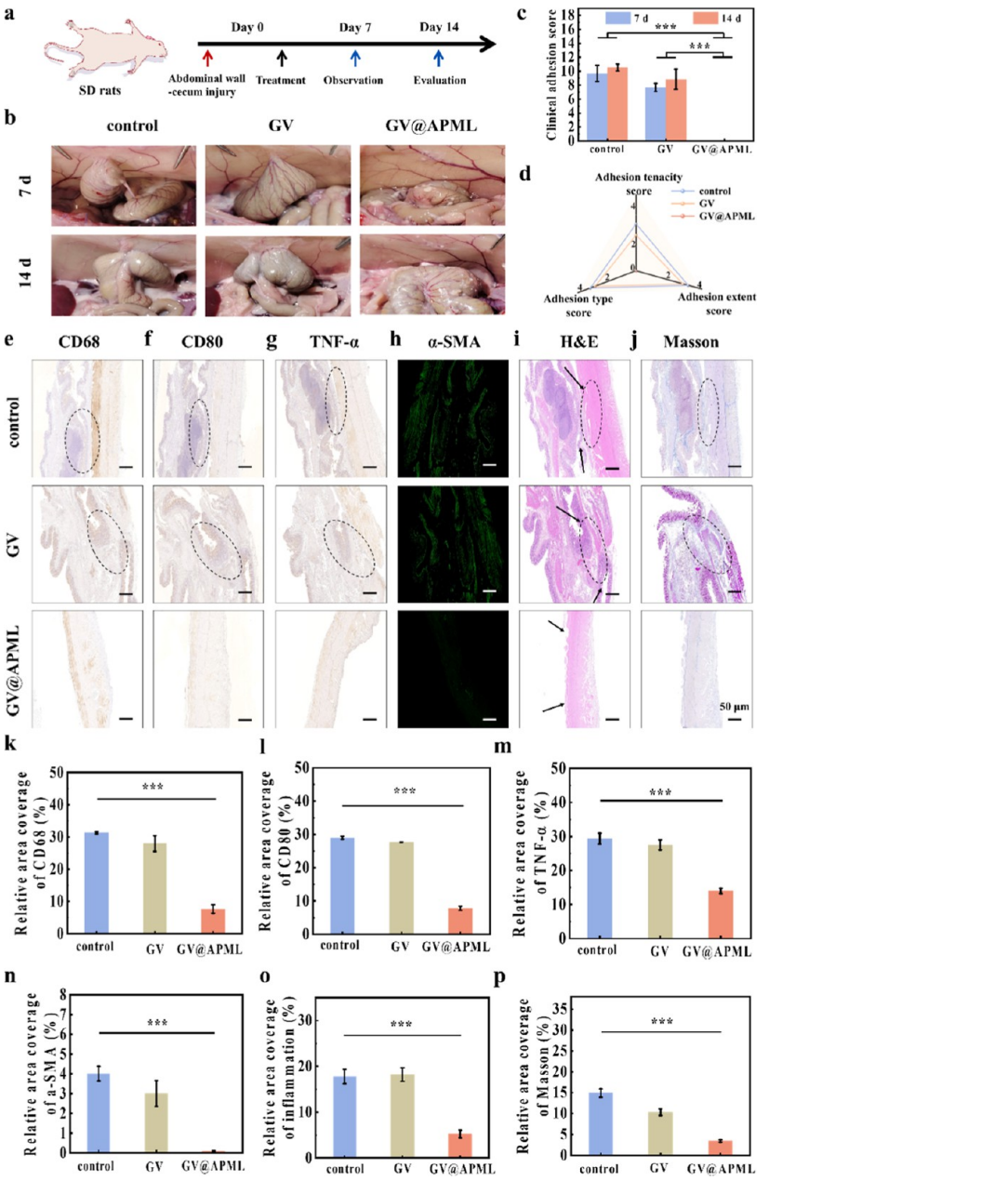


Figure 5. (a) Schematic diagram of the process to prevent PA with the use of the composite hydrogel. (b) Representative photos of postoperative adhesions in the different treatments on the seventh and 14th day. (c) Quantitative scoring analysis of total PA and (d) quantitative scoring analysis of adhesion extent, adhesion type, and adhesion tenacity of PA areas with different treatments at day 14 after operation. Histomorphological evaluation of GV@APML hydrogel in antiadhesion in vivo. Immunohistochemical staining images of CD68 (e), CD80 (f), TNF-α (g). (h) Immunofluorescence of α-SMA. (i) H&E staining images. (j) Masson staining images. Quantitative analysis of CD68 (k), CD80 (l), TNF-α (m), α-SMA (n) and (o) quantitative analysis of inflammation and (p) collagen density with different treatments at day 14 after treatment (the site of adhesion was circled).

(as the control), GV gel (only γ-PGA cross-linked with the PVA hydrogel), GV@APZ gel (only APZ nanocarrier integrated within the GV hydrogel), and GV@APML (encapsulated AMP and PL based on GV@APZ gel). The wound conditions of different groups were photographed on day 3 (Figure 4b). To detect the antibacterial effect of samples in vivo, the infected wound tissues were collected on day 3 and the number of bacteria was estimated using a colony formation assay. As shown in Figure 4c,d, the antibacterial ratios of the GV and GV@APZ hydrogel groups were 43% and 48%, respectively, indicating the inherent bacterial inhibition property of GV hydrogel. In contrast, the sterilization efficiency of the GV@APML hydrogel loaded with AMP was significantly improved such that more than 97.3% of the bactericidal ratio had been achieved. This result indicated the remarkable antibacterial performance of the composite GV@APML hydrogel in vivo which could match the results of antibacterial evaluation in vitro, and further demonstrated that the bioactive APML nanoparticles could effectively release the bactericidal agents in the infectious microenvironment. Furthermore, to preliminarily investigate the anti-inflammatory activity of the composite hydrogels in the infectious microenvironment, the harvested tissues were stained for histological and immunochemical analysis. As displayed in Figure 4e, much

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more inflammatory cells including macrophage and neutrophils recruited at sites that processed with PBS, GV, and GV@APZ hydrogel, clarifying that infection-induced inflammatory responses persisted in the wound. While, after treatment with the GV@APML composite hydrogel, which could release AMP in an intelligent way, the inflammatory response was significantly reduced, displaying only a mild inflammatory response. To further verify the anti-inflammatory effect of the composite hydrogel, the representative biomarkers such as CD68 (an indicator of macrophages) and IL-6 (a proinflammatory chemokine) were evaluated in detail through immunohistochemical staining. As shown in Figure 4f,h, the groups of GV and GV@APZ demonstrated detectable reduction for the expression of CD68 compared with the control group, which may be relevant to their moderate bacterial inhibition effect as proved in the antibacterial assessment. By contrast, the GV@APZ group presented the lowest expression of CD68 compared with the other three groups, consistent with the above histological analysis result that the lowest number of inflammatory cells recruited in the infected tissue was treated with the GV@APML composite hydrogel. Interestingly, as evaluating the level of proinflammatory IL-6, there was no obvious difference between control and GV hydrogel groups, while mild decrease was observed in the GV@APZ group (Figure 4g,i), which might be attributed to the ROS-scavenging ability of the PDA integrated in the GV@APZ. In addition, after treatment with the GV@APML hydrogel, the expression of the IL-6 was dramatically lower than that in the other groups. This excellent anti-inflammatory ability of the GV@APML hydrogel might result from the synergy of the existing PDA acting with an anti-inflammatory effect and the rapid bactericidal properties of the AMP loaded within the APML nanoparticles. These results collectively signified that GV@APML composite hydrogel had strong in vivo antibacterial activity to prevent infection and suppress inflammation.

3.10. In Vivo Antiadhesion Efficacy of Hydrogels. At present, traditional methods for preventing PA, such as devices based on physical barriers or drug treatments alone, achieve unsatisfactory outcomes due to the inherent imperfection of these treatments as well as the diverse pathogenesis of PA formation. Hence, we hypothesize that the construction of an "all-in-one" composite hydrogel capable of physical barrier capability and pharmacological effects may be an ideal strategy to completely inhibit the occurrence of PA. In this context, a full-thickness abdominal wall injury with 10 mm in diameter and abrasion of cecum were constructed on a rat model to evaluate the adhesion prevention effect of the constructed GV@APML composite hydrogel in vivo (Figure 5a). After the cecum and abdominal wall were injured, the application of samples (GV or GV@APML) on both traumatized surfaces was immediately executed. The group treated with only PBS was used as the negative control. The level of abdominal visceral adhesion formation was examined on the seventh and 14th days after operation. Before dissection of adhesion, the cecum and abdominal wall tissues were collected. The corresponding blinded graders were given to comprehensively evaluate the degree of adhesion according to the clinical classification of the extent, type, and tenacity of adhesion (Table S1).[57,58] As shown in Figures 5b–d and S9, the adhesion between the abdominal wall and the cecum was most severe in the control group, which had high adhesion levels (10.2) regarding the adhesion type (3.7), adhesion extent (3.7), and adhesion tenacity (2.8) on day 14 postsurgery. The GV hydrogel group exhibited a lower degree of adhesion (~8.9) than the control group. By contrast, the GV@APML hydrogel was the most effective treatment in preventing adhesions and all the adhesion scores were distinctly lower than those of the GV hydrogel. Taken all together, these results meant that GV hydrogel has almost no antiadhesion efficiency when used alone and the addition of APML dramatically improved the antiadhesion capability of the hydrogel.

After injury, abnormal activation of immune cells, such as M1-type macrophages, and the release of cytokines may lead to an imbalance between local inflammation and repair of injured tissues, which further aggravates the degree of PA.[59] In this context, the pro-inflammatory M1 macrophage-related markers (such as CD68 and CD80) and representative pro-inflammatory cytokine (such as TNF-α) were evaluated through immunohistochemical staining and corresponding qualitative analysis to further study the effects of the hydrogels on these inflammatory factors. The levels of CD68+ and CD80+ macrophages were much lower in the GV@APML composite hydrogel group than in the control and GV hydrogel groups at day 7 (Figure S10a,b,g,h) and day 14 (Figure 5e,f,k,l). In addition, the TNF-α which was rapidly and abundantly expressed after stimulation with inflammatory factors was also greatly reduced in the GV@APML group on the seventh (Figure S10c,i) and 14th day (Figure 5g,m). Thus, the lowest accumulation of M1 and expression of TNF-α indicated that GV@APML composite hydrogel could significantly inhibit the inflammatory response caused by the abdominal wall injury and cecum abrasion. Meanwhile, the imbalance between fibrinogenesis and fibrinolysis is a key feature of PA formation. To investigate the activated state of fibroblast, the expression of α-SMA which is an important fibroblast marker and would be up-regulated in abdominal adhesion sites was assessed through immunofluorescence staining.[6] Compared with the control group, the expression of α-SMA was modestly reduced in the GV hydrogel group on the seventh (Figure S10d,j) and 14th day (Figure 5h,n), indicating that the hydrogel alone possessed inherent antiadhesion property by means of the physical barrier effect. Notably, the GV@APML hydrogel-treated group presented no detectable signals of α-SMA, demonstrating that there was almost no adhesion developed at all. This satisfactory outcome might be related to two aspects: the physical barrier effect derived from the GV hydrogel and the excellent inhibiting property of the hybrid BN integrated in the hydrogel for inflammatory responses, especially for the TNF-α which is instrumental in maintaining the equilibrium of fibrinolytic system and prevent adhesion formation. Furthermore, H&E staining and Masson trichrome staining were conducted for histologically visualizing wound repair status and tissue adhesion around the injury sites. On day 7 postsurgery (Figure S10e,k), a large number of inflammatory cells were observed between the cecum and the muscle layer of the injured abdominal wall in the control (without treatment) and GV hydrogel group (without incorporation of BNs). As shown in Figure 5i,o, after day 14 of the surgery, the negative control group and the GV hydrogel group still exhibited severe tissue fusing with the accumulation of abundant inflammatory cells between the cecum and abdominal wall muscle layer, confirming that the physical barrier alone was not enough to prevent the formation of PA as above proved. In contrast, a distinct isolation of abdominal wall with complete reconstructed epithelial structures presented on the surface of the

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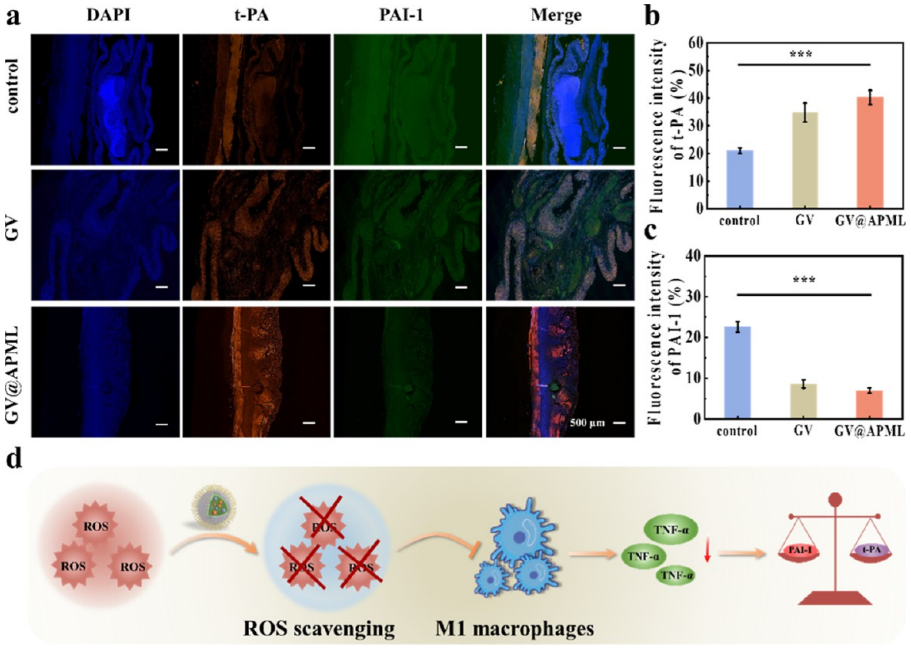


Figure 6. Influence of GV@APML hydrogel on fibrinolytic activity. (a) Immunofluorescence staining of different groups at 14 days. Tissue sections were stained with DAPI (blue), t-PA-Alexa Fluor 594 (red), and PAI-1-Alexa Fluor 488 (green). Semiquantitative analysis of the t-PA (b) and PAI-1 (c). (d) Mechanisms of GV@APML hydrogel for postoperative antiadhesion.

injured sites in the GV@APML groups, demonstrating the satisfactory antiadhesion and wound pro-healing performances of the constructed composite hydrogels, which might be closely related to the integration of PDA and the loaded PL in the composite hydrogel. In addition, as displayed in Figure S10f,l (on days seventh postsurgery) and Figure 5j,p (on days 14th postsurgery), the Masson trichrome staining further confirmed the H&E staining results that connective/adhesive regions with excessive deposition of collagen (a key feature of PA formation) could be easily identified in the control and GV hydrogel groups, while these phenomena could be well inhibited in the composite hydrogel-treated groups. These results collectively indicated that the GV@APML composite hydrogel simultaneously integrating physical antiadhesion with excellent anti-inflammation had superior potential in completely preventing adhesion formation.

3.11. Potential Mechanism of the GV@APML Hydrogel against PA. The imbalance between fiber degradation and deposition was one of the most crucial causes inducing the formation of postoperative adhesions as the previous reports.[6] t-PA and plasminogen activator inhibitor-1 (PAI-1) are important factors regulating the fibrinolytic activity, whose imbalance will result in the abnormal relationship between fiber degradation and deposition.[38] Thus, the effect of GV@APML hydrogel on protein expressions of t-PA and PAI-1 around the wound site was further investigated by double immunofluorescence staining. On the 14th day, the fluorescence images of PAI-1 and t-PA in the damaged tissues of each group were shown in Figure 6a. The control group (without any treatment) had the highest PAI-1 (green fluorescence) and the lowest t-PA (red fluorescence) at 2 weeks after surgery. This phenomenon might account for the above results in that the control group presented the most severe PA. After GV hydrogel treatment, the expression of PAI-1 reduced to a certain degree, which was consistent with the GV hydrogel treatment alone still providing a minor inhibiting capability for the formation of PA taking advantage of the inherent physical barrier effect. In sharp contrast, the group treated with GV@APML composite hydrogel exhibited the lowest PAI-1 and highest t-PA expressions (Figure 6b,c), suggesting that GV@APML composite hydrogel participated in the regulation of the dynamic balance between PAI-1 and t-PA which were key regulatory factors in preventing postoperative adhesions (Figure 6d).

4. CONCLUSIONS

In summary, an inflammation-responsive injectable hydrogel (GV) was first designed and developed via the formation of phenylboronic ester bonds between the PGA−PBA and PVA under physiological conditions. The dynamic borate ester bond endowed the hydrogel with excellent injectability, self-healing, and biodegradability, which made it feasible to cover irregular wounds and acted as a physical barrier for initially preventing PA. Further, a BN, APML, was elaborately designed and constructed with multifunction including specific recognition for the activated macrophages, prolonged accumulation in inflammatory sites, and remarkable anti-inflammatory, antibacterial, and pro-healing effects. After the BNs were integrated with the designed inflammation-responsive hydrogel, a composite GV@APML hydrogel was successfully prepared, which could sustainably release the BNs in the inflammatory microenvironmental wound defect to perform the above biofunctions. It is worth mentioning that combining the physical barrier effect of the GV and the multiple biofunctions of the BNs, the GV@APML composite hydrogel demonstrated a remarkable antiadhesion effect in a clinically relevant cecum-abdominal wall adhesion model. Notably, the pathological analysis emphasized that the immunoregulatory ability of the GV@APML for balancing the fiber degradation and deposition through regulating the expression of PAI-1 and t-PA might be a significant passway accounting for their super antiadhesion property. In addition, subcutaneous implantation

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ACS Appl. Mater. Interfaces 2024, 16, 60933−60947

test in vivo demonstrated its excellent biocompatibility. Overall, this study provides a universal and efficient tactic for the design and construction of multifunctional hydrogels combining the physical and pharmacological synergy to safely, effectively, and conveniently prevent postoperative adhesions.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsami.4c10238.

Materials and reagents; clinical adhesion score criteria; SEM image of the nanoparticles; ^{1}H NMR spectra, FT-IR spectra, UV−vis spectrum of γ-PGA, PBA, and PGA−PBA; optical images of the self-healing process of hydrogel; degradation profiles of GV hydrogel; H&E staining of the muscle tissues from mice on days 7; TEM images of macrophages incubated with PZ and APZ; additional DPPH scavenging efficiency for control and $HZrO_2$; the morphological images of L929 cells by cytoskeleton staining; additional quantitative scoring analysis; and histomorphological evaluation of GV@APML hydrogel in antiadhesion in vivo on the seventh day (PDF)

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Notes

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